

Take-Home Assignment: File Storage & Sharing

We don't expect a production system — we care about your judgement, code quality, and how you communicate trade-offs. Please don't spend more than a weekend on this.

Use a coding agent. We don't want you hand-typing boilerplate. The world is moving agentic, so please build this *with* a coding agent of your choice (Claude Code, Cursor, Codex, Copilot, Aider, etc.). What we're evaluating is how well you direct the agent, review and correct its output, and make the decisions it can't make for you. Tell us in the README which agent(s) you used and how you worked with them.

1. The Problem

A small team needs a simple place to keep and share documents. They want to:

- Upload a document and keep it somewhere safe.
- Share a document with someone outside the team by sending them a link.
- Create a **workspace**, invite colleagues into it, and organize documents there so the team isn't passing files around one at a time.

Build a small full-stack application (API + web UI) that does this.

This brief is intentionally incomplete. Where details are missing — what a "link" can do, who is allowed to do what inside a workspace, what happens when things expire or get deleted, how invitations behave for people who don't have an account yet — **you decide**. Make the call a reasonable product team would make, and write it down.

2. Constraints

Language	Python or Node.js (TypeScript preferred)
Database	PostgreSQL , with migrations
File storage	Not in PostgreSQL. Use an S3-compatible store (MinIO locally) or local disk behind an abstraction you could swap.
UI	In scope. Any framework or server-rendered templates. Functional over pretty.
Run	<code>docker-compose up</code> brings up everything.
Tests	Some. Aim them at the parts that would embarrass you if they broke.

Anything else — auth approach, frameworks, file size limits, role model, API shape — is yours to choose.

3. Open Space: Product Improvement

Once the core works, pick **one** thing you believe a real user of this product would want next and either build it or write a one-page design note for it. Some directions people take: versioning, password-protected links, quotas, folders, audit trail, virus scanning, search, notifications. Yours can be anything. Tell us why you chose it.

4. Deliverables

1. A Git repository with source, migrations, tests, `docker-compose.yml`, and a `.env.example`.
2. A **README** that covers:
 - How to run it (target: under 5 minutes from clone).
 - A short architecture overview.
 - **Assumptions and decisions.** Every gap in this brief that you filled — what you chose and why. This is the most important section.
 - Security considerations you addressed, and ones you knowingly left.

- Your product improvement (built or designed).
 - How you worked with your coding agent: what you delegated, where it went wrong, how you caught it.
 - What you'd do next with more time.
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5. What We're Looking For

- **Problem framing** — Did you notice what was missing and resolve it sensibly, or did you build only what was literally written?
- **Security and authorization** — Can one user reach another user's files? Are share links guessable? Is the storage backend exposed?
- **Data modelling** — Does the schema reflect the domain, and does it handle deletion and membership cleanly?
- **Code quality** — Readable, organized, idiomatic. Storage and auth aren't tangled into request handlers.
- **UI** — The core flows work end-to-end in a browser, and the interface reflects permissions correctly.
- **Communication** — The README explains *why*. Trade-offs are named honestly.
- **Agent direction** — You steered the agent rather than accepting its first output, and you can explain every part of the codebase. Expect the follow-up interview to include a code walkthrough.

We do **not** grade on visual design, cloud deployment, or test coverage percentage.

6. Questions

You can email us with clarifying questions, but you won't be penalized for making a reasonable assumption on your own and documenting it — that's part of the exercise.